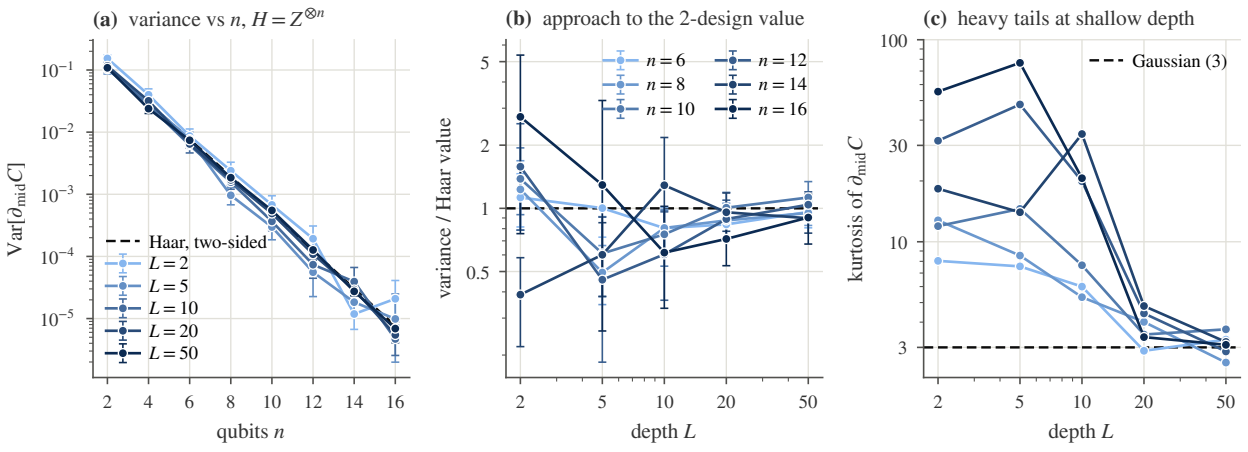
(a) variance vs n, $H = Z^{\otimes n}$
$\mathrm{Var}[\partial_{\mathrm{mid}} C]$
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
Haar, two-sided
$L = 2$
$L = 5$
$L = 10$
$L = 20$
$L = 50$
2 4 6 8 10 12 14 16
qubits n
(b) approach to the 2-design value
variance / Haar value
5
2
1
0.5
$n = 6$
$n = 8$
$n = 10$
$n = 12$
$n = 14$
$n = 16$
2 5 10 20 50
depth L
(c) heavy tails at shallow depth
kurtosis of $\partial_{\mathrm{mid}} C$
100
30
10
3
Gaussian (3)
2 5 10 20 50
depth L